

GULLFAKS C, Norway

WMO: 013000

Lat: 61.204N

Lon: 2.269E

Elev: 78

StdP: 100.39

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-0.2	0.8	-8.4	1.9	1.5	-6.7	2.1	2.2	21.6	4.5	20.1	4.7	7.7	330	0.907

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	2.2	18.7	15.3	16.8	14.5	15.7	14.0	16.1	17.6	15.1	16.3	14.3	15.2	5.7	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.4	11.0	16.5	14.6	10.5	15.6	13.8	10.0	14.8	45.2	17.6	42.7	16.4	40.5	15.2	19.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.3	16.9	15.3	DB	-2.0	21.2	1.3	2.3	-3.0	22.9	-3.8	24.3	-4.5	25.6	-5.5	27.3
			WB	-3.5	16.7	1.4	1.6	-4.5	17.8	-5.4	18.7	-6.2	19.6	-7.2	20.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.5	5.0	4.8	5.3	6.6	8.3	10.8	13.0	13.5	11.9	9.3	7.1	5.8
	DBStd	3.59	1.87	1.96	1.77	1.74	2.04	1.75	2.01	1.63	1.75	1.83	2.08	2.16
	HDD10.0	869	154	144	146	102	59	9	1	0	4	34	87	130
	HDD18.3	3595	413	378	404	351	309	227	166	149	192	281	336	388
	CDD10.0	318	0	0	0	1	8	32	94	109	62	12	1	0
	CDD18.3	2	0	0	0	0	0	0	2	1	0	0	0	0
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
(14) Wind	WSAvg	8.0	9.8	9.1	8.4	7.5	6.9	6.6	6.4	6.5	7.8	8.5	9.0	9.4
(15) Precipitation	PrecAvg	1341	161	132	127	72	48	65	81	100	117	141	139	156
	PrecMax	1652	252	257	237	123	83	132	153	162	190	216	220	240
	PrecMin	954	76	58	46	26	22	26	40	66	48	58	43	65
	PrecStd	165	48	53	46	23	16	24	22	25	34	41	39	42
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.0	9.2	12.2	16.6	18.9	21.9	20.3	17.5	13.5	11.2	10.2
		MCWB	8.4	8.0	7.7	8.0	11.5	13.4	16.2	15.6	14.1	11.2	10.1	9.4
	2%	DB	8.7	8.4	8.5	10.3	13.9	16.0	19.6	17.7	15.7	12.6	10.7	9.6
		MCWB	7.8	7.3	7.3	7.6	10.0	12.6	16.0	15.5	13.4	11.1	9.6	8.8
	5%	DB	8.2	8.0	8.0	9.3	12.0	13.7	17.0	16.5	14.8	12.1	10.2	9.1
		MCWB	7.2	6.9	6.9	7.4	9.6	11.6	14.9	14.8	13.1	10.7	9.2	8.2
	10%	DB	7.8	7.5	7.6	8.7	10.8	12.6	15.5	15.5	14.1	11.6	9.8	8.7
		MCWB	6.6	6.3	6.3	7.2	9.1	11.0	14.3	14.1	12.9	10.2	8.7	7.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.9	8.6	8.6	9.1	12.6	15.0	17.5	16.9	14.8	12.5	10.9	9.9
		MCDB	9.0	8.7	8.8	10.0	15.6	17.5	20.5	18.3	16.1	12.7	11.1	10.0
	2%	WB	8.3	8.0	8.0	8.6	11.0	12.9	16.2	16.1	14.1	11.7	10.2	9.2
		MCDB	8.5	8.2	8.2	9.3	12.8	15.0	18.1	17.4	14.9	12.1	10.4	9.4
	5%	WB	7.6	7.4	7.3	8.1	10.0	12.1	15.4	15.2	13.7	11.2	9.6	8.6
		MCDB	8.1	7.8	7.7	8.8	11.3	13.1	16.9	16.0	14.4	11.7	10.0	9.0
	10%	WB	6.9	6.6	6.6	7.6	9.3	11.5	14.4	14.4	13.2	10.6	9.0	7.9
		MCDB	7.6	7.3	7.3	8.4	10.3	12.4	15.4	15.2	13.8	11.3	9.6	8.5
(35) Mean Daily Temperature Range	5% DB	MDBR	2.6	2.5	2.5	2.5	2.2	2.2	2.2	2.2	2.2	2.2	2.4	2.6
		MCDBR	2.7	2.7	2.5	3.1	4.3	3.8	4.3	3.5	2.9	2.4	2.4	2.8
	5% WB	MCWBR	3.5	3.1	2.8	2.2	2.6	2.4	2.3	2.3	2.1	2.5	3.0	3.6
		MCWBR	2.7	2.6	2.4	2.5	3.5	2.9	3.6	3.0	2.5	2.2	2.5	2.8
(40) Clear-Sky Solar Irradiance	taub	taub	0.325	0.349	0.363	0.377	0.365	0.358	0.375	0.367	0.361	0.341	0.313	0.382
		taud	2.070	2.203	2.285	2.314	2.422	2.479	2.451	2.471	2.441	2.321	2.134	2.329
	Ebn at Noon	Ebn at Noon	354	604	746	813	858	871	843	821	752	614	369	221
		Edh at Noon	43	70	90	105	101	98	98	89	77	61	39	28
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.19	0.68	1.83	3.59	5.03	5.19	4.68	3.66	2.21	1.03	0.30	0.12
	RadStd	RadStd	0.02	0.10	0.25	0.35	0.34	0.58	0.38	0.31	0.18	0.14	0.04	0.01

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (20 neighbors)		N/A	+0.65	N/A	N/A	N/A	N/A	-83	-118	N/A	N/A	

Nomenclature: See separate page